

# Substitution

Section 4.5

11/4/25

Ex1)

$$\int 2x(x^2+5)^3 = \frac{du}{dx} = 2x$$

$u = x^2 + 5$   
 $du = 2x dx$

$$\int u^3 du = \frac{1}{4} u^4 + C$$

$$= \frac{1}{4} (x^2+5)^4 + C$$

Ex2)

$$\frac{1}{3} \int x^2 \sqrt{x^3+1} dx$$

$u = x^3 + 1$   
 $\frac{du}{dx} = 3x^2$   
 $du = 3x^2 dx$

$$\frac{1}{3} \int u^{1/2} du = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} + C$$

Ex3)

$$\int \frac{3x}{(5+x^2)^2} =$$

$u = 5+x^2$   
 $du = 2x dx$

$$\frac{3}{2} \int u^{-2} du$$

$$-\frac{3}{2} u^{-1} + C = \frac{-3}{2(5+x^2)} + C$$

$$\int_0^2 x \sqrt{4+x^2} dx =$$

$u = 4+x^2$   
 $du = 2x dx$

$$\frac{1}{2} \int_4^8 u^{1/2} du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} \Big|_4^8$$

$$\frac{1}{3} (8^{3/2} - 4^{3/2})$$